Glossary: words, symbols and small examples

Use a definition to reconstruct an idea, then return to the chapter where it is developed

> **A DEFINITION IS THE BEGINNING OF UNDERSTANDING**
>
> For each term, explain the small example and invent a changed example. Follow the chapter link for assumptions, derivations and practice. A familiar symbol can have different local meanings; the notation conventions and formula sheet make those meanings explicit.

Mathematical language

Quantifier

“For every” and “there exists” specify the scope of a statement. “For each state there is a safe action” allows feedback; “there is one action safe for every state” demands a fixed action. These orders need not be equivalent. Primer 0.

Assumption

A condition under which a conclusion is asserted. If a sensor error is bounded by 0.1, an interval certificate using that bound makes no claim for error 0.2. Checking the algebra and checking the assumption are different tasks. Proof techniques.

Function

A rule assigning one output to each allowed input. A controller $u = \pi(x)$ maps a measured or estimated state to an action; its domain and output units are part of its definition. Functions.

Invariant set

A set that the dynamics keep a trajectory inside, when it starts inside and the stated disturbance/control conditions hold. For $x_{k+1} = x_k/2$, $[-1, 1]$ is invariant. Invariance alone does not say whether a target is reached. Safety meanings.

Fixed point

An input left unchanged by a map: $F(x) = x$. For $F(x) = 0.5x + 0.07$, the fixed point is 0.14, not zero. A persistent error can therefore change the steady state. Fixed points.

Supremum and maximum

The supremum is the least upper bound; a maximum must also be attained. The set $[0, 1)$ has supremum 1 and no maximum. Writing an optimizer requires an existence argument, not only an upper bound. Supremum and infimum.

Counterexample

One valid instance that contradicts a universal claim. A system that fails from one allowed initial state disproves “safe for all initial states”; many successful simulations cannot repair that universal statement. Proofs and disproofs.

Linear algebra and geometry

Norm

A measure of vector size with homogeneity, the triangle inequality and zero only at the zero vector. For $(3, 4)$, the Euclidean norm is 5, the 1-norm is 7 and the infinity-norm is 4. A perturbation radius must identify which size it uses. Norms.

Dual norm

The norm that bounds a linear functional over the chosen unit ball. For $a = (2, -1)$, a Euclidean perturbation of radius r changes $a^\top x$ by at most $r\sqrt{5}$; a coordinate-wise radius r changes it by at most $3r$. Dual norms.

Operator norm

The largest amplification ratio of a linear map for the specified input and output norms. The Euclidean operator norm of $\mathrm{diag}(2, 1)$ is 2, even though some input directions are amplified by only 1. Matrix norms.

Positive semidefinite matrix

A real symmetric matrix P with $v^\top P v \geq 0$ for every real vector v. Zero in some nonzero direction is allowed; positive definiteness excludes it. $\mathrm{diag}(1, 0)$ is semidefinite and singular. Quadratic forms.

Singular value decomposition

A factorization into orthonormal coordinate changes and nonnegative stretch factors. The largest singular value gives the Euclidean operator norm. A finite power-iteration estimate can underestimate that value and cannot automatically replace a certified upper bound. SVD.

Conditioning

Sensitivity of a mathematical problem's solution to changes in its data. Nearly parallel sensor equations may determine a state uniquely while making its reconstruction highly sensitive to noise. Uniqueness is not the same as accurate recovery. Conditioning and pseudoinverses.

Optimization

Gradient

The vector describing a differentiable function's first-order change. For $f(x) = x_1^2 + 2x_2^2$, it is $(2x_1, 4x_2)$. It predicts sufficiently small changes; a large step needs more than the linear prediction. Gradients.

Convexity

A function lies below the line segment joining any two points on its graph over a convex domain. For differentiable convex functions, first-order supporting inequalities help certify global optimality. A stationary point of a nonconvex function need not be a global minimum. Convexity.

Feasible point

A point satisfying every constraint. A robot action within its velocity limit may still fail its clearance constraint. A good objective value cannot compensate for infeasibility unless the problem explicitly defines such a trade-off. Constraints.

Projection

A point in a set closest to the proposed point under a specified distance. Projecting nominal command -0.8 onto $[-0.5, 0.8]$ gives -0.5. Changing the distance or feasible set can change the result. Action filtering.

Lagrange multiplier

A nonnegative price on an inequality constraint in a chosen sign convention. In an appropriate sensitivity setting it measures how the optimal value changes when a budget is relaxed. It is neither a probability nor an automatic physical safety controller. Constrained choices.

KKT conditions

Conditions joining stationarity, primal and dual feasibility, and complementary slackness. With the appropriate convexity assumptions they can certify an optimum; necessity requires suitable regularity. Checking only stationarity omits the constraints. Duality and KKT.

Trust region

A region in which an update model is considered reliable or a change is limited. An ellipsoid $s^\top Hs \leq r^2$ weights directions through H. A trust-region constraint by itself does not prove a nonlinear safety requirement. Policy updates.

Probability and evidence

Expectation

A probability-weighted average. A binary failure indicator has expectation equal to its failure probability, while a discounted sum of such indicators represents another quantity. A small expected cumulative cost need not forbid every failure trajectory. Random variables.

Independence

A property permitting joint probabilities to factor into marginal probabilities for the specified events or variables. Two readings from the same unmodeled sensor bias need not be independent. Repetition alone does not justify multiplying probabilities. Conditioning.

Confidence statement

A statement whose probability refers to the randomness of a procedure's data or experiment. The source of randomness and sampling assumptions determine its meaning. A confidence level does not turn an estimated error bound into a deterministic bound. Concentration.

Union bound

The probability that any event occurs is at most the sum of their probabilities. It needs no independence. Four failure probabilities each bounded by 0.1 give union probability at most 0.4, hence joint success at least 0.6. Joint guarantees.

Conditional value at risk

A tail-risk statistic, conventionally the mean of the worst $(1 - \alpha)$ probability mass. Atoms may contribute only part of their mass at the threshold; simply averaging every outcome above a quantile can be wrong for a discrete distribution. Risk measures.

Exchangeability

A joint distribution is unchanged by permuting the observations. It underlies rank-based conformal calibration; independent identically distributed data are one sufficient setting. A time-varying deployment environment need not be exchangeable with past calibration data. Conformal prediction.

Marginal coverage

A coverage probability averaged over the specified calibration and test randomness. It does not automatically give the same coverage conditional on every subgroup or a simultaneous guarantee for a whole future trajectory. Calibration project.

Systems and control

State

The variables a model uses to predict future evolution from current information and inputs. If future admissible actions depend on a remaining budget, physical position alone may be an incomplete decision state. State-space models.

Feedback

Choosing an input from observed or estimated system information. A pump action can depend on tank volume; one constant action may not handle both an empty and a full tank. Measurements, delay and actuator capability belong in the model. Tank project.

Lyapunov function

A function measuring departure from an equilibrium whose evolution helps establish stability or convergence under stated conditions. For $x_{k+1} = x_k/2$, $V = x^2$ decreases by $3x^2/4$. Persistent disturbances change what decrease can prove. Stability and error tubes.

Control barrier function

A function representing a safe set whose derivative condition constrains admissible inputs, under the required regularity and feasibility assumptions. A nonempty action set and correct treatment of sampling are part of using the certificate. Clearance filter.

Viability

The possibility of keeping a state inside a required set by suitable future actions. A state can be inside the set yet already unable to avoid future failure, for example when its braking capability is insufficient. Viability kernels.

Model predictive control

Repeatedly optimize a finite future plan, execute its first action, update the state information and plan again. Feasibility of today's finite plan is not automatically recursive feasibility; terminal conditions and robust margins can supply the missing argument. MPC.

Invariant error tube

A set bounding the deviation between actual and nominal trajectories. For $q_{k+1} = 0.5q_k + e_k$, $|e_k| \le 0.07$, radius 0.14 is invariant. Tightening a nominal state bound by that radius reserves room for the deviation. Tube calculation.

Learning and certificates

Policy

A decision rule in a sequential model. A stationary Markov policy uses the current modeled state, while more general policies may depend on history. A mixture choosing fast mode with probability p is a policy in the one-state example. MDPs.

Bellman equation

An equation splitting a value into an immediate contribution and a discounted next-state value. In a one-state constant-reward model, $V = r + \gamma V$ yields $V = r/(1-\gamma)$ for $0 \le \gamma < 1$. Value functions.

Gaussian process

A probabilistic function model specified by a mean and covariance kernel. A posterior mean and variance depend on model assumptions; a small posterior variance is not by itself a frequentist safety guarantee for an arbitrary unknown function. Kernels and uncertainty.

RKHS norm

A kernel-dependent measure of an entire function in a reproducing-kernel Hilbert space. Bounding it restricts the function class. It is not the same as checking a finite list of function values or bounding the largest observed slope. Function spaces.

Lipschitz bound

A uniform inequality $|f(x) - f(y)| \leq L\|x - y\|$ on a specified domain. It transfers a known margin to nearby points. Sampled differences alone do not generally establish an upper bound on L. Safe tuning project.

Certificate

Evidence establishing a precisely stated property under explicit assumptions. A global output bound, an invariant-set inequality and a statistical coverage result certify different properties. The intended decision determines which is useful. Comparing guarantees.

Formal verification

Checking a mathematical statement in a precise formal system. A Lean theorem verifies its encoded claim and premises, not every informal interpretation or empirical assumption beside it. The project's coverage records keep that correspondence explicit. Evidence conventions.

Read the symbols in context

In a dynamical model, x commonly denotes a state and u an input; in optimization, x may be a generic decision vector. π can denote a policy. γ commonly discounts future contributions. δ may be a confidence failure budget, a perturbation or a trust-region radius: these are not interchangeable because the glyph is the same. A Lipschitz constant L may equal zero; formulas dividing by it need a separate case.

Use a four-line notation card for a difficult derivation: symbol, type/dimension, units or norm, and where it is defined. For example, "$u \in \mathbb{R}$, litres per sample, chosen pump change" prevents accidentally treating it as a flow rate in litres per second. "$\delta \in \mathbb{R}^2$, normalized feature perturbation, infinity-norm bound 0.2" prevents silently substituting a Euclidean ball.

To test understanding, take one formula from a chapter and translate every quantifier into words. Then change either its norm, its uncertainty set or its horizon and explain why the old conclusion may no longer apply. Return to the reading path when the local definition is clear.